Block - 1
Aptitude (Time & work)

① Given

A works = 10 days

B works = 15 days

A and B Together = ?

1 Day of A = 1/10

1 Day of B = 1/15

$$A \& B = \frac{1}{10} + \frac{1}{15} = \frac{\cancel{25}\,5}{\cancel{150}\,30} = 1/6$$

Total time = 1/6 $\boxed{6 \text{ Days}}$

2) A, B, C = 6, 8, 12 Days

A = 6, B = 8, C = 12

$$A + B + C = \frac{1}{6} + \frac{1}{8} + \frac{1}{12}$$

$$= \frac{4 + 3 + 2}{24}$$

2	6	8	12
2	3	4	6
2	3	2	3
3	3	1	3
	1	1	1

$= 2 \times 2 \times 2 \times 3$

LCM - 24

$$1 \text{ Day} = \frac{\cancel{9}}{\cancel{24}} = \frac{3}{8}$$

$$\frac{1}{3/8} = 8/3 = 2\frac{2}{3} \text{ days}$$

$= 2.67$ days

A work - 12 days after 4 days B joing and finish 4 Days

$A = 1/12$

$4 \times 1/12 = 1/3$

$1 - 1/3 = 2/3$

$\frac{2/3}{4} = \frac{\cancel{2}}{\cancel{12}} = 1/6$

$B = (A+B) - A$

$= \frac{1 \times 2}{6 \times 2} - \frac{1}{12}$

$= \frac{2-1}{12} = 1/12$

$\boxed{12 \text{ Days}}$

5 work - 8 days

? work - 4 days

$5 \times 8 = x \times 4$

$40 = x \times 4$

$\frac{40}{4} = x$

$10 = x$

$\boxed{10 \text{ work}}$

5) Tank fills - 6 hours

Emptied - 8 hours

$$= \frac{1}{a} - \frac{1}{b}$$

$$= \frac{1}{6} - \frac{1}{8}$$

$$= \frac{8-6}{48}$$

$$= \frac{2}{48} = \frac{1}{24}$$

$$\boxed{24 \text{ Hours}}$$